P 3.5 Process Validation and Evaluation/Report

Enclosed Herewith

PROCESS VALIDATION PROTOCOL

FOR
GENERIC NAME:

SOLIFENACIN SUCCINATE TABLETS 5 MG

Protocol No.	LLS/T/PVP/005/001	**Mother Batch Size**	120000 Tablets
Export Stability batch Size (Alu/ALU Blister Pack)	3000 Tablets		
Dosage form	Oral Solid Dosage	**Effective date**	14/02/2021
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2.0 PROTOCOL APPROVAL

Activity	Name	Department	Designation	Signature & Date
Prepared By	Ankita Karale	Quality Assurance	Officer	14/02/2021
Checked By	Gajanan Ingole	Quality Assurance	Sr. Executive	14/02/2021
	Shahrukh Khan	Quality Control	Executive	14/02/2021
	Abhijeet Mandekar	Production	Sr. Executive	14/02/2021
Approved By	Taher Sayyed	Quality Assurance	Manager	14/02/2021

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3.0 OBJECTIVE

The Objective of this protocol is to validate the manufacturing process for Solifenacin Succinate Tablets 5 mg and to establish that the process consistently gives the product that meets all the predetermined specifications and quality attributes.

4.0 SCOPE

The protocol is applicable for the manufacturing of Solifenacin Succinate Tablets 5 mg manufactured for a standard batch size of 120000 Tablets at different stages; Sifting, Dry Mixing, Wet granulation, Drying Lubrication, Compression , coating process and blister packing of 3000 tablets. Three consecutive batches will be considered for process validation

5.0 RESPONSIBILITY

Sr. No.	Department	Responsibility
1.	Quality Assurance	To prepare Validation protocol, To monitor all Validation activities To prepare Validation report.
2.	Store	To arrange raw & packing material and to maintain finished goods stock.
3.	Production	To perform all Validation activities as per Protocol. To monitor Validation procedure
4.	Manager Production	To review process Validation protocol.
5.	Quality Control	To analyze the samples collected as per specification and to prepare an analysis report.
6.	Manager QC	To review Validation protocol
7.	Engineering	To monitor the machine performance and environmental controls. To provide required utilities for manufacturing.
8.	Microbiology	To monitor Validation activities, To monitor all microbiological test procedures.
9.	Manager Quality Assurance	Approval of Validation protocol and authorize all Validation activities

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6.0 PRODUCT DETAILS

Generic name	Solifenacin Succinate Tablets 5 mg
Dosage form	Oral Tablets
Composition	**Each Film Coated Tablet contains:** Solifenacin Succinate BP 5 mg Excipients q.s.
Shelf life	36 months from the month of manufacturing.
Mother Batch Size	18.000 Kg
Batch quantity	120000 Tablets
Therapeutic category	Antimuscarinics
Description	Light yellow coloured, round shaped, film coated tablet, plain on both sides.
Weight of Tablet	153.00 mg (Coated)
Packing	ALU/ALU Blister of 10 Tablets
Storage condition	Store below 30^{0}C, in dark & dry place.

7.0 TYPE OF VALIDATION: Concurrent Validation

REASON FOR VALIDATION: New product Introduction

8.0 TRAINING DETAILS:

- The Validation team shall be authorized by Head- QA or Designee.
- All the personnel involved in the manufacturing and packing of Validation batches. Sampling and testing of sample should be appropriate trained both in their job related activities and on the process Validation protocol by Head-QA or Designee.

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9.0 VALIDATION CRITERIA

- Product shall be manufactured and packed as per as per BMR and BPR.
- Three batches shall be considered for Validation study.
- Current version of standard operating procedure to be followed.
- All activities shall be carries out as per protocol.
- Record the observations stage wise in the respective batch manufacturing and packing record
- Record the yield after Blending, Compression, Coating and packing in manufacturing and packing record.
- The data generated during manufacturing and captured in the executed batch manufacturing records is compiled and verified against the acceptance criteria defined in the process Validation protocol.
- Three optimum batches shall be subjected to stability study
- Validation report shall be prepared based on the analytical reports / observation of Validation batch.

10.0 CRITERIA FOR REVALIDATION

- The process shall be revalidated whenever there is change in;
- Change in site of manufacturing
- Change in batch size.
- Change in Equipment
- Manufacturing process change.
- Manufacturing Formula (Change in composition or components)
- Change in API vendor or critical excipient.
- Change in specification of input material.
- Abnormal trend in quality parameters identified in APQR.
- Trend of Out of specification.

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11.0 MANUFACTURING FORMULA

Mother Batch Size: 120000 Tablets

Ingredients	Item code	Grade	Category	Qty. / Batch in kg	Qty. / Tablet in mg
DRY MIX					
Solifenacin Succinate	A036BP01A	BP	Active Substance	*0.600	5.000
Lactose Monohydrate	E101BP	BP	Diluent	12.300	102.500
Maize Starch	E041BP	BP	Bulking agent	#4.800	40.000
BINDER					
Hypromellose	E033BP	BP	Binder	0.120	1.000
Purified water	NA	--	Solvent	@2.458	Q.S.
LUBRICATION					
Magnesium stearate	E040BP	BP	Lubricant	0.180	1.50
Core tablet weight					150.00
Film Coating					
Readycoat Iron Oxide Yellow 36FO193220	E015IH	IH	Film former	**0.468	3.000
Purified water	NA	--	Solvent	@4.212	Q.S.
Total Tablet weight					153.00

Note*: Actual quantity of Solifenacin Succinate to be calculated on the basis of Assay (dried basis) and % LOD.

Compensation will be with corn starch in actual batch.

**30% extra qty. to avoid the loss during coating

@Quantity not remains in final product.

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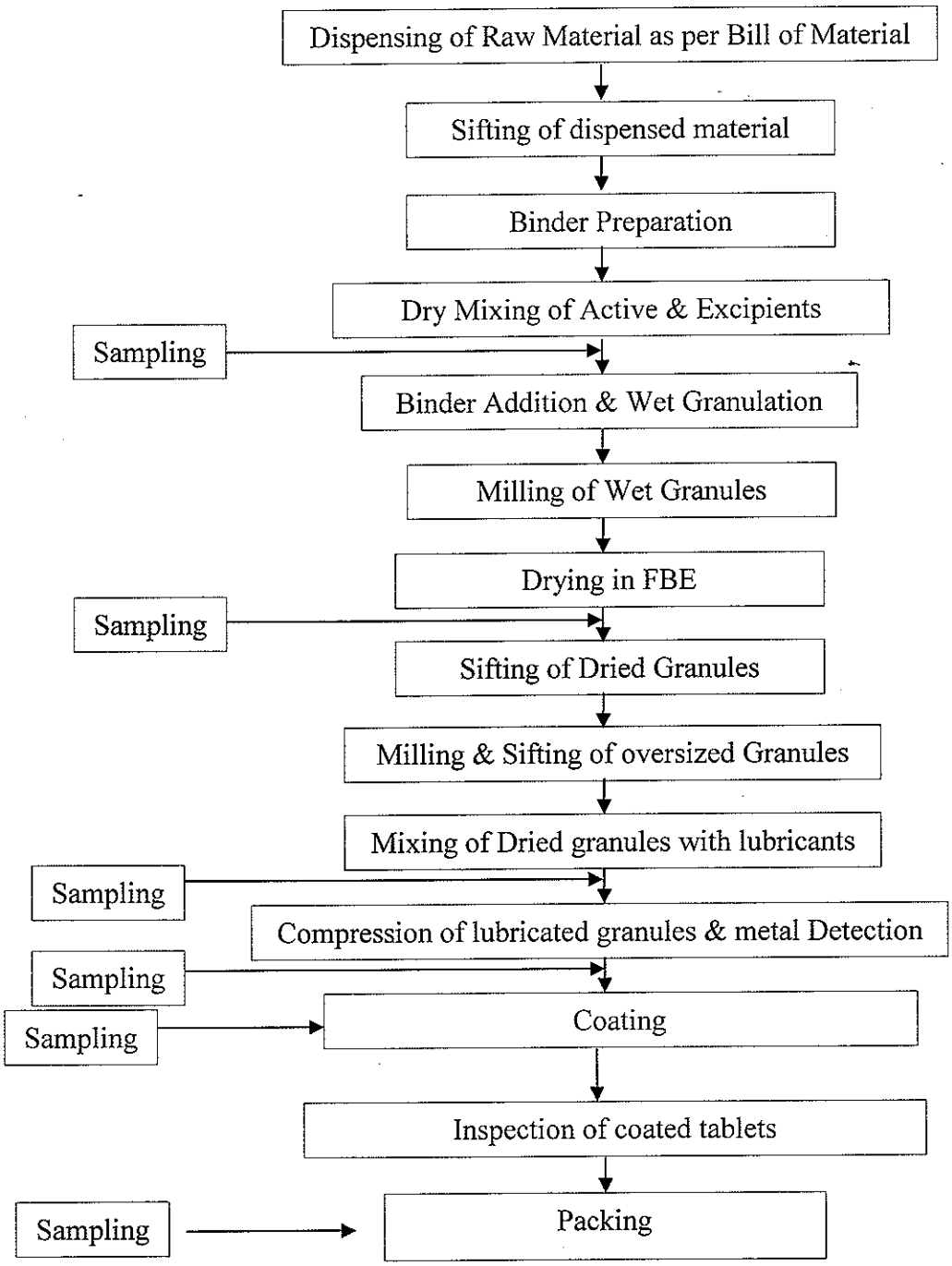
12.0 REFERENCES DOCUMENTS

Master Formula Record	MFR/RD/T-05/00
Master packing Record	MPR/RD/T-05/00
Batch Manufacturing Record	BMR/T-05/00
Batch Packing Record	BPR/T-05/00

Note: All current versions of relevant documents shall be referred

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13.0 PROCESS FLOW CHART:



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14.0 DESCRIPTION OF EQUIPMENT, FACILITY INVOLVED IN MANUFACTURING PROCESS

- All the major equipments and instrument used for process & facility as listed below shall be verified for Installation, Operational and Performance Qualification and recorded.

Sr. No.	Name of the equipment	Equipment ID
Warehouse		
01	OFCB	EQ/WRH/OFCB/003, EQ/WRH/OFCB/004
02	Weighing Balance	INQ/WRH/WEB/001
03	Weighing Balance	INQ/WRH/WEB/002
04	Weighing Balance	INQ/WRH/WEB/003
05	Weighing Balance	INQ/WRH/WEB/004
Production		
01	Double Deck Vibratory Sifter (30 inch)	EQ/OSD/VSF/002
02	High Shear Mixer Granulator - 100L	EQ/OSD/HSMG/001
03	Pillar Type Bin Blender - 300 L	EQ/OSD/PBB/001
04	Binder Preparation Vessel 60 L	EQ/OSD/BPV/001
05	Fluid Bed Equipment (FBE 125 C)	EQ/OSD/FBE/001
06	Tap Cone Mill -100	EQ/OSD/TCM/001
07	27 Station 'D' Tooling Double Rotary Compression Machine	EQ/OSD/COM/001
08	Combo metal detector and de-duster	EQ/OSD/DMD/001 EQ/OSD/DMD/002
09	Capsule and tablet inspection belt	EQ/OSD/CTIC/001
10	Blister Packing Machine	EQ/OSD/RBPM/001
11	Packing Conveyor Belt	EQ/OSD/PCB/001
12	Weighing Balance	INQ/OSD/WEB/003
13	Weighing Balance	INQ/OSD/WEB/002
14	Weighing Balance	INQ/OSD/WEB/004
15	Weighing Balance	INQ/OSD/WEB/005
16	Weighing Balance	INQ/OSD/WEB/007
17	Weighing Balance	INQ/OSD/WEB/013
18	Weighing Balance	INQ/OSD/WEB/014

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Sr. No.	Name of the equipment	Equipment ID
IPQA		
01	Weighing Balance	INQ/IPQA/WEB/001
02	Vernier Caliper	INQ/IPQA/VC/001
03	Disntegration Test Apparatus	INQ/IPQA/DT/001
04	Leak Test Apparatus	INQ/IPQA/LT/001
05	Hardness Tester	INQ/IPQA/THT/001
06	Halogen Moisture Analyzer	INQ/IPQA/HMA/001
07	Friability Test apparatus	INQ/IPQA/FT/001
Quality Control		
01	Analytical Weighing Balance	INQ/QCD/WEB/002, INQ/QCD/WEB/003
02	HPLC	INQ/QCD/HPLC/001,INQ/QCD/HPLC/002 INQ/QCD/HPLC/006.
03	UV Visible Spectrophotometer	INQ/QCD/UV/001
04	IR	INQ/QCD/IR/001
05	Vernier Caliper	INQ/QCD/VC/002
06	Disintegration test Apparatus	INQ/QCD/DT/001
07	Friability Test apparatus	INQ/QCD/FT/001
08	Hardness Tester	INQ/QCD/THT/001
09	pH meter	INQ/QCD/PH/001
10	Tap Density Apparatus	INQ/QCD/TAP/001
11	Dissolution test apparats	INQ/QCD/DIS/001, INQ/QCD/DIS/002
12	Autotitrator	INQ/QCD/AT/001
13	UV Cabinet	INQ/QCD/UVC/001
14	Stability Chember	EQ/QCD/STB/001 to EQ/QCD/STB/003
Microbiology		
01	Weighing Balance	INQ/QCM/WEB/001, INQ/QCM/WEB/002
02	HLAF	EQ/QCM/HLAF/001
03	Vertical Autoclave	EQ/VA/001
04	Biosafety Cabinet	EQ/QCM/BSC/001

Note: All the manufacturing equipment and instrument qualification status is reviewed and found within validity period. All calibrated instrument to be used.

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15.0 LIST OF APPROVED VENDORS:

Sr. No	Name of Raw Material & Grade	Approved Vendor Name & Address
Raw Material (API and Excipients)		
01	SOLIFENACIN SUCCINATE BP	**Amoli Organics Pvt.Ltd.** 407, Dalmal House, Jamnalal Bajaj Road, Nariman Point, Nariman Point, Mumbai, Maharashtra 400021
02	LACTOSE MONOHYDRATE BP	**Prachin Chemicals** 1307/6-7, Phase IV, G.I.D.C., Naroda, Ahmedabad-382330, Gujarat, India.
03	MAIZE STARCH BP	**Shivangan Food and Pharma Products Pvt. Ltd.** Survey No. 226/1, Rawal Industrial Estate, Shahada Road, Dadanagar,Dondaicha-425408 Dist: Dhule (M.S.)
04	HYPROMELLOSE BP	**LOTTE Fine Chemical co., Ltd.** 47, namdong-daero 79beon-gil namdong-gu, Incheon 21700, Republic of Korea
05	MAGNESIUM STEARATE BP	**Nitika Pharmaceutical Specialities Pvt. Ltd.** Block No.03 of Plot No. 1296 1297/08, Ward No.03, Mouza-Waddhamana, Tah-Hingna, Amravati Road, Nagpur-440023.
06	READYCOAT IRON OXIDE YELLOW 36FO193220	**Dhara Life Science Pvt. Ltd. (Unit-I)** Block No.1273, At. Moti Bhoyan, Vadsar-Kalol .Road, Ta. Kalol, Dist: Gandhinagar-382721, Gujarat, India.

Sr. No	Packing Material	Approved Vendor Name & Address
01	AF Solifenacin T 5 mg 224 MM	**Shah's Super Seals India Pvt.Ltd.** W-B-24 M.I.D.C., Ambad, Near Nashik Merchant Bank, Nashik-422 010
02	BASE FOIL 226 MM	**Indian Cork Mills Pvt. Ltd.** S.No.57/3(1), Bhenslore Dunetha, Nani Daman 396210
03	Mono Carton (105x75x13mm)	**H.K.Printers Pvt Ltd** H.K House ,inside Neelkanth near sai baba temple ,satavali vasai,Palghar, Mumbai
04	Pack Insert Solifenacin T 5 mg	**Kirti Printers** Near Akola Urban Bank, old cotton market, Akola-444001

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16.0 PROCESS VALIDATION METHODOLOGY

- During the course of Validation the documentation systems, laboratory controls, raw materials, final product of Solifenacin Succinate Tablets 5 mg will be evaluated.
- The Process Validation Methodology will consist of three basic parts: process parameters monitoring, the routine in process and final product release testing & additional sampling and testing.
- The Validation exercise for one batch of production scale will be done to assess the process consistency and the individual batches should have raw material (API) of different Manufacturing batch number.
- The Validation protocol requires pre-approval for sampling plan and acceptance criteria before execution.
- At the conclusion of the process Validation, a capability study will be performed to determine whether the process is consistent and capable of repeatedly and reliably producing a finished product of the specified quality.
- Operate the manufacturing & packing process as per the SOP.

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17.0 BRIEF DISCRIPTION OF MANUFACTURING PROCEDURE

Instructions:

- Wear personal protective ancillaries as and when required during all stages of manufacturing.
- All dispensed materials shall be cross checked before use for weights and labeling.
- Check the area and equipment cleanliness as per SOP before starting the activity and record.
- During product to product change over line clearance to be verified by QA.
- Proper labeling shall be maintained during the manufacturing process.
- Operate the equipment's as per the SOPs.
- Record the temperature (22 ± 3 °C) and relative humidity (%) 50 ± 10 during granulation, blending, Compression, Coating and inspection.
- Preserve all the raw material & API labels along with Batch Manufacturing after completion of blending

Sr. No.	Manufacturing stage	Stage	Specification	Test method
1.	Before starting	Environment control	Temperature(°C): within 22 ± 3 (RH%) : 50 ± 10	Digital Hygrometer
		Cleanliness of equipment & utensils	Should be clean	Visual inspection

17.1 SAMPLING PROCEDURE AT DIFFERENT STAGES:

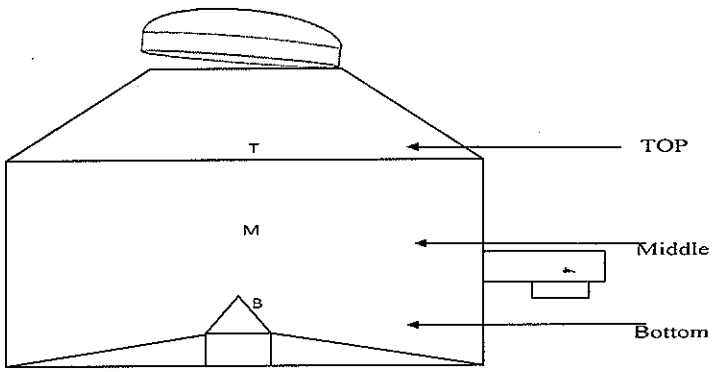
Dry Mixing:

- This step involves mixing of sifted API and Excipients in HSMG with impeller slow speed for 10 minutes with chopper off.
- Samples shall be drawn from HSMG after mixing for 10 minutes
- Sampling to be done by sampler rod using appropriate die.
- The sample quantity to be withdrawn from top, middle and bottom of the HSMG i.e 5gm each. Sample shall be withdrawn in duplicate quantity. Samples of the Top, middle and bottom shall be mix to make composite. 5 g composite sample shall be send to quality control for analysis.
- Collect composite samples in glass vials. Duplicate composite samples shall be kept in Quality assurance. The duplicate set of sample shall be destroyed after the evaluation of the results of the samples submitted in Quality control. Collect the sample of each lot.

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- Label the polybag with details like Product name, Batch No. Batch Size, Mfg. date, Exp. date Stage, sample quantity and sampled by and submit the samples to Quality control for analysis.

Drawing-I

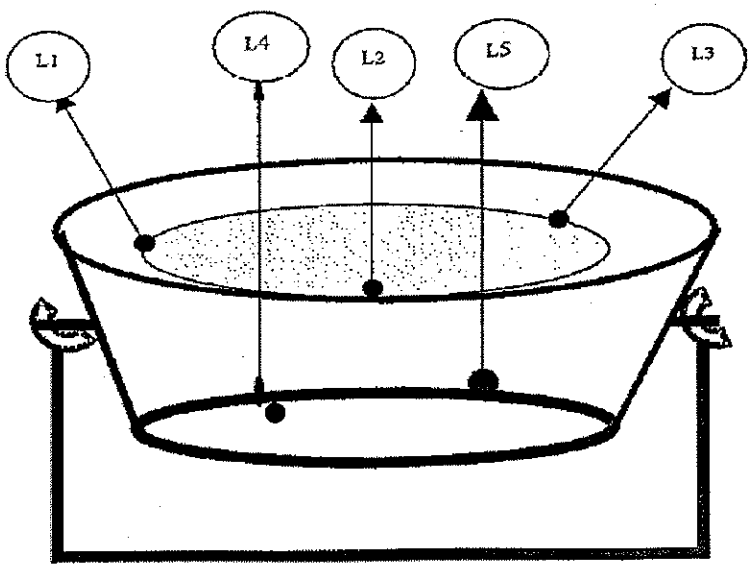


- **Drying:**

The drying of granules shall be performed in the Fluid bed Equipment.
The batch to be loaded for drying & target temperature shall be uniform and drying shall be performed till the LOD is achieved as per temperature mention in BMR
Collect the samples from the locations shown in below diagram make composite and check the LOD.
The LOD should be NMT 2 %.

Drawing-II

Sampling Points at FBD Bowl



Location 1: Rear Left

Location 2: Front Middle

Location 3: Rear Right

Location 4: Bottom Left

Location 5: Bottom Right

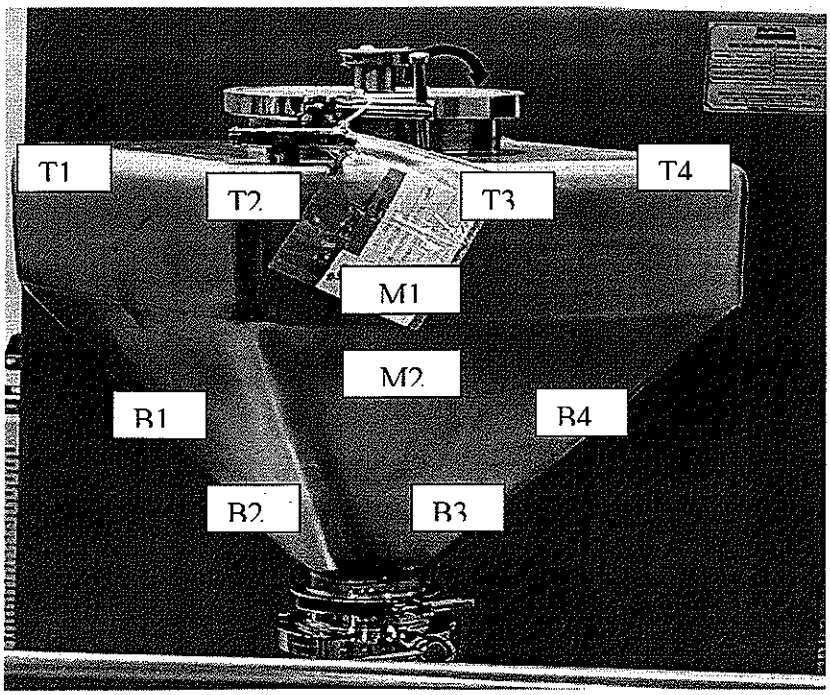
- **Lubrication:**

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- Load sized granules into pillar bean blender and mixed it with pre-sifted magnesium stearate
- Mix the magnesium stearate for 5 minutes at 8 RPM by operating the blender.
- Samples to be withdrawn from 10 locations from different location shown in Diagram III. 820 mg to 2460 mg samples shall be collected from individual locations in vials. Location number and product details shall be specified on each vials.
- Collect 55 gm approx. as Composite sample from top, middle and bottom.
- Sampling to be done with sampling rod using appropriate die.
- Collect composite samples into self-sealing poly bag, Label the polybag with details like Product name, Batch No. Batch Size, Mfg. date, Exp. date Stage, sample quantity and sampled by and submit the samples to Quality control for analysis.

Stage: Lubrication Equipment: Pillar Type Bin Blender

Drawing-III



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Compression:

Compression activity to be carried out as per instructions mentioned in batch manufacturing record.

Set the machine for different speeds at Minimum, Optimum and Maximum RPM and check the physical parameters like Description, Average Weight, Group Weight of 20 Tablets, Uniformity of Weight, Disintegration time, Hardness, Friability, Thickness, Uniformity of Dosage Unit (Mass Variation), Assay,

Collect the sample of lower & higher hardness for Description, Average Weight, Group Weight of 20 Tablets, Uniformity of Weight, and Disintegration time, Hardness, Friability, Thickness, Assay and Dissolution.

Collect samples for Description, Group Weight of 20 Tablets, Uniformity of Weight, Disintegration time, Hardness, Friability, Thickness, Uniformity of dosage units , Assay, during initial, middle and end of compression cycles.

Perform the Description, Average mass, Uniformity of Weight, Disintegration time, Friability, Assay, Water Content on composite sample

After Coating composite sample withdrawn for Disintegration time, Thickness, Group Weight of 20 Tablets, appearance, Dissolution and assay.

Acceptance criteria for compression stage :

PRODUCT PARAMETERS	SPECIFICATION
Appearance	White to off white, round shaped tablet plain on both sides.
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)
Average Weight of 20 tablets	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)
Thickness (mm)	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)

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Hardness	NLT 3.0 kp
Friability	Not more than 1.0 % w/w
Disintegration Time	Not more than 15 minutes

Remarks: RPM of Compression machine shall be finalized after the validation batches.

COATING:

- **Film Coating Solution preparation: (For each lot)**

STEP NO.	MANUFACTURING PROCEDURE		OBSERVATION
1	Take the 4.212 kg purified water in solution preparation vessel		
2	Add Readycoat Iron Oxide Yellow 36FO193220 in purified water under stirring in solution prepartion vessel till clear solution	**Start Time**	
		End Time	
		Stirrer RPM	
3	Check the integrity of S.S. SCREEN before filter operation.	**Integrity**	
4	After getting the solution, filter the solution through S.S. SCREEN	**Start Time**	
		End Time	
5	Verify integrity of S.S SCREEN after filter operation	**Integrity**	

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Gun Validation:

Nozzle No.	RPM	Time		Tare wt.	Gross wt.	Net wt.	Spray rate gm / minute
		From	To				
1.							
2.							
3.							

Inprocess Observation:

Controlled Variables	Acceptance Criteria	Batch No.	Batch No.	Batch No.
Inlet Air temp	60^0C to 80^0C *			
Exhaust Air Temp.	45^0C to 60°C *			
Pan RPM	1-5 RPM*			
Weight Gain	2.0 %			
Peristaltic pump speed(RPM)	1 to 3RPM			

Acceptance criteria:

Parameters	Acceptance criteria/Limits
Appearance	Light yellow coloured, round shaped, film coated tablet, plain on both sides.
Weight of 20 coated tablets	3.06 gm ± 3.0 % (2.168 gm to 3.152 gm)
Average weight of coated tablet	153.00 mg ± 7.5% (141.52 mg to 164.47 mg)
Thickness (mm)	3.28 mm ± 0.3 mm (2.98 mm to 3.58 mm)
Disintegration time	NMT 30 min

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PRIMARY PACKING:

- **Process Parameters:** Monitor the following parameters and document
  - Sealing Temperature
  - Sealing Quality
  - Knurling Quality
  - Leak test
  - Speed of Machine

Challenge Test	Procedure	Acceptance Criteria	Actual Observation		
			Batch No	Batch No	Batch No
Alu. Plain Foil & Alu. Plain Foil end	End of Alu. Plain Foil and Alu. Plain Foil	Machine stops at zero position			
Alu. Plain Foil & Alu. Plain Foil splice detection	Splice detector shall detect and reject the blister	The blister with splice should get rejected at rejection station			
Rejection of blister under forming and sealing station	After restart of machine the blister under forming and sealing station should be rejected	The blister get rejected at rejection station			
Leak test	Perform the leak test using leak test apparatus Qty-	No blister pocket should be leak			

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Test	Specification	Actual Observation		
		Batch No	Batch No	Batch No
Cavity Forming Temperature	_____To_____ °C To be validated			
Sealing Temperature	_____To_____ °C To be validated			
Cutting	Should be proper (Found OK/Not OK)			
Knurling	Should be proper (Found OK/Not OK)			
Speed of Machine	____ To_____ Blister /min.			
Leak Test	Blister Quantity			
	Leak test (Pass/Fail)			

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CRITICAL PROCESS PARAMETERS:

Manufacturing stage	Machine Parameters	Process Parameters	CQA (Critical Quality Attributes)
Dry Mixing	1.Speed of impeller	Mixing time	1. Assay
Wet Granulation (Preparation of wet granules)	1.Speed of impeller 2.Chopper speed	1.Binder addition time 2.Mixing time	1. Impeller speed: Slow RPM and Fast RPM. 2. Chopper speed: Fast RPM. 3. Impeller Amperage Load
Drying	1. Inlet temperature 2. Outlet temperature	1. Drying time	1.Moisture content
Milling of oversized dried granules	1.Screen size	1.Milling Speed	1. Consistent dried granules
Lubrication	1. Speed of blender	1.Mixing time	1. Assay 2. Water Content 3. TD and BD
Compression	1.Speed of Compression Machine 2. Force feeder speed 3. Tonnage	Low hardness High hardness Minimum Speed Optimum Speed Maximum Speed	Average weight Group Weight Thickness Hardness DT Friability Assay Water content

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Manufacturing stage	Machine Parameters	Process Parameters	CQA (Critical Quality Attributes)
Coating	1. Speed of pan. 2. Peristaltic pump RPM	1. Inlet air temp. 2. Bed Temperature. 3. Exhaust Temperature. 4. Spray rate.	Description, Weight of 20 enteric coated tablets, Thickness, Disintegration test Assay Water content Average weight Dissolution
Blister packing	1.Forming temp. Sealing roller temp. 2. Speed of blister machine	-	Leak Test

Note: In Compression stage challenge test, IPQA shall Verified only.

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PROCESSING STAGE, SAMPLING PLAN, TEST PARAMETERS & ACCEPTANCE CRITERIA:

Processing Stage	Sampling Plan	Test parameter Done by Production	Test parameter Done by IPQA	Test parameter Done by QC	Acceptance criteria
Dry mixing	**Refer Schematic Drawing-I** Samples to be drawn from Top, middle and bottom each location 5 gm Composite 5gm (Withdraw Duplicate sample) (40 gm)	--	--	1. Description 2. Assay.	**Description:** White to off white Powder. Assay : 90.0 % to 110.0%
Drying	**Refer Schematic Drawing-II** Collect the samples from the locations shown in diagram make composite sample of 5g by mixing samples of individual location and check the LOD. (Check the LOD of each lot.)	1. Moisture content (LOD)	Verify By	--	NMT 2%
Lubrication	**Refer Drawing-III** 1.5gm samples shall be collected from individual locations in vials. Location: Top (T1, T2, T3 & T4) Middle (M1,&M2) Bottom (B1,B2,B3 & B4) and Composite 55.0 gm	Moisture content (LOD)	Moisture content (LOD)	BU sample: Description, Assay, Composite: Water Content, Tap density & BD	**Description**: White to off white granules. **Water** NMT 5% **Bulk Density**: For Information **Tapped Density:** For Information **Assay :** 90.0 % to 110.0%

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Processing Stage	Sampling Plan	Test parameter Done by Production	Test parameter Done by IPQA	Test parameter Done by QC	Acceptance criteria
Compression	QC: samples For Machine Speed: At Minimum RPM At Maximum RPM At Optimum RPM 40 tablets Composite (RHS + LHS) Withdraw Duplicate sample **(240 Tablets)**	Appearance, Average Weight, Group Weight of 20 tablets, Thickness, Hardness, Friability, Disintegration Time	Verified By IPQA	1. Description 2. Assay 3. Average weight	To comply as per specification
	Low & High Hardness 40 tablets Composite (RHS + LHS) Withdraw Duplicate sample **(240 Tablets)**	Appearance, Average Weight, Group Weight of 20 tablets, Thickness, Hardness, Friability, Disintegration Time	Verified By IPQA	1. Description 2. Assay 3. Average weight	
	Initial Compression, Middle Compression End Compression 40 tablets Composite (RHS + LHS) Withdraw Duplicate sample **(240 Tablets)**	Appearance, Group Weight of 20 tablets, Thickness, Hardness, Friability, DT	Verified By IPQA	1. Description 2. Assay 3. Average weight	
	40 Tabs composite.	--	--	Description Assay Average weight Water content	
Coating	Composite Sample:50 tablets	Appearance Weight of 20 enteric coated tablets Average weight of	Appearance Weight of 20 enteric coated tablets Average	Description, Assay, Water content, Dissolution.	To comply as per specification

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Processing Stage	Sampling Plan	Test parameter Done by Production	Test parameter Done by IPQA	Test parameter Done by QC	Acceptance criteria
		enteric coated tablet Thickness	weight of enteric coated tablet Thickness		
Primary Packing	Machine Speed: Sealing & Forming Temperature Challenge	Knurling	Verified By IPQA	--	Should be proper
		Leak Test	Verified By IPQA	--	All the Blister packs should be 100% leak free.
		Blister Cutting	Verified By IPQA	Related substance	1.Vertical and horizontal cuttings should be visually acceptable in shape and size
Packing (secondary & tertiary packing)	Sampled Quantity Finished sample 150 Tablets for QC. Analysis and 30 Tablets micro. Analysis Reserve Sample :360 Tablets Stability Sample: Accelerated =510 Tablets Long Term = 990 Tablets Intermediate = 990 Tablets	NA	NA	NA	Comply as per packed product specification.

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PROCESS VALIDATION PROTOCOL

Generic Name	Solifenacin Succinate Tablets 5 mg	**Protocol No.**	LLS/T/PVP/005/001
Supersedes	NIL	**Page No.**	27 of 31

Processing Stage	Sampling Plan	Test parameter Done by Production	Test parameter Done by IPQA	Test parameter Done by QC	Acceptance criteria

Note: In Compression stage and in primary Packing challenge test, IPQA shall Verified only.

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18.0 Finished Product Analysis Report:

Sr. No.	Tests	Specifications	Batch No.	Batch No.	Batch No.
1.	**Description**	Light yellow coloured, round shaped, film coated tablet, plain on both sides. Such 10 tablets packed in ALU/ALU Blister.			
2.	**Identification by IR**	The infrared absorption spectrum of the dried residue is concordant with the reference spectrum of Solifenacin succinate RS/WS.			
3.	**Average weight**	153.00 mg ± 7.5% (141.53 mg to 164.48 mg)			
4.	**Uniformity of weight**	Not More than 2 tablets deviate from average weight by 7.5% None Tablet deviat from the averag weight by more tha 15%			
5.	**Content Uniformity**	AV NMT 15			
6.	**Hardness**	NLT 3 Kp			
7.	**Friability**	NMT 1.0% w/w			

Sr. No.	Tests	Specifications	Batch No.	Batch No.	Batch No.
8.	**Disintegration time**	NMT 30 minutes			
9.	**Dissolution**	NLT 75% (Q) of the labeled amount			
10.	**Assay**	95.0% to 105.0% of the stated amount			
11.	**Microbial Limit Test**	TAMC: 1000 cfu/gm. TYMC: 100 cfu/gm. Escherichia coli: absent/gm. Salmonella: absent/10 gm. P. aeruginosa: absent/gm. S. aureus: absent/gm.			

19.0 Re-Validation:

In case of the requirements for Validation, the Validation of the critical steps shall be Undertake through addendum protocol to this protocol or a separate protocol.

20.0 Deviation and OOS:

Deviation and OOS observed in Validation activity shall be documented as per standard procedure and noted down in the qualification report.

21.0 Change Control:

Any change in process, procedure and specification in Validation activity shall be documented as per standard procedure and noted down in the Validation report

22.0 Summary Report:

Prepare a summary report based on the report of the findings stage wise

23.0 Conclusion:

Based on the Validation study and summary report a conclusion for the process verification should be prepared and based on the conclusion the process verification batches should be subjected to stability study.

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24.0 ABBREVATIONS

LLS	:	Leben Life sciences
PVP	:	Process Validation Protocol
Mfg.	:	Manufacturing
Lic.	:	License
No.	:	Number
API	:	Active Pharmaceutical Ingredient
AR. No.	:	Analytical Report Number
Sr. No.	:	Serial Number
HPLC	:	High Performance Liquid Chromatography
IR	:	Infra-Red
°C	:	Degree Celsius
%	:	Percent
LOD	:	Loss On Drying
FP	:	Finished Product
SOP	:	Standard Operating Procedure
QA	:	Quality Assurance
RH	:	Relative Humidity
Temp.	:	Temperature
BOM	:	Bill of Material
Min.	:	Minutes
MM	:	Millimeter
RPM	:	Rotation Per Minute
IPQA	:	In process Quality Assurance
Hrs.	:	Hours
NMT	:	Not More Than
KG	:	Kilogram
DT	:	Disintegration Time
QC	:	Quality Control
Av. Wt.	:	Average Weight
Tab	:	Tablet
NLT	:	Not Less Than

PROCESS VALIDATION REPORT

FOR

GENERIC NAME: SOLIFENACIN SUCCINATE TABLETS 5 MG

Report No.	LLS/T/PVR/005/001	**Ref. Protocol No.**	LLS/T/PVP/005/001
Mother Batch Size	120000 Tablets	Export Stability Batch Size (ALU/ALU Blisters)	3000 Tablets
Dosage form	Oral Solid Dosage	**Report Effective date:**	28/03/2021

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2.0 REPORT APPROVAL

Activity	Name	Department	Designation	Signature & Date
Prepared By	Bhagwat Kankatao	Quality Assurance	Sr. officer	27/03/2021
Checked By	Gajanan Ingole	Quality Assurance	Sr. Executive	27/03/2021
	Sruttish Bhandurge	Quality Control	Manager	27/03/2021
	Sachin Sakhare	Production	Manager	27/03/2021
Approved By	Taher Saiyed	Quality Assurance	Manager	28/03/2021

3.0 OBJECTIVE

The Objective of this report is to validate the manufacturing process for Solifenacin Succinate Tablets 5 Mg to establish that the process consistently gives the product that meets all the predetermined specifications and quality attributes.

4.0 SCOPE

The report is applicable for the manufacturing of Solifenacin Succinate Tablets 5 Mg manufactured for a standard batch size of 1,20,000 Tablets at different stages; Sifting, Dry Mixing, Wet granulation, Drying, Sizing, Lubrication, Compression, coating & packing of 3000 tablets. Three consecutive batches were manufactured successfully at Leben Life Sciences Pvt. Ltd., Akola for process validation.

5.0 RESPONSIBILITY

Sr. No.	Department	Responsibility
1.	Quality Assurance	Compiling and analyzing data and test results and preparing the final report. To ensure that the processing of the batch is carried as per the approved BMR. To review the completed batch records and report.
2.	Production	To assist in collection of data. To fill the batch records.
3.	Manager Production	To review process validation report.
4.	Quality Control	To analyze the samples collected as per specification and to prepare an analysis report.
5.	Manager QC	To review validation report
6.	Manager Quality Assurance	To review and approve validation report.

6.0 Product Details:

Generic name	Solifenacin Succinate Tablets 5 Mg
Dosage form	Oral Tablets
Composition	**Each Film Coated Tablet contains:** Solifenacin Succinate BP 5 mg Excipients q.s.
Shelf life	36 months from the month of manufacturing.
Mother Batch Size	18.000 Kg
Mother Batch quantity	120000 Tablets
Therapeutic category	Antimuscarinics
Description	Light yellow coloured, round shaped, film coated tablets plain on both sides.
Weight of Tablet	153.00 mg (Coated)
Packing	ALU/ALU Blister of 10 Tablets
Storage condition	Store below 30^{0}C, in dark & dry place.

6.1 Details of mother batch taken for process validation:

Product Details	Batch No.	Mfg. Date	Exp. Date	Mother Batch Size
Solifenacin Succinate Tablets 5 Mg	LT0101001	03/2021	02/2024	120000 Tablets
	LT0101002	03/2021	02/2024	120000 Tablets
	LT0101003	03/2021	02/2024	120000 Tablets

7.0 TYPE OF VALIDATION: Concurrent Validation

8.0 TRAINING DETAILS:

- The Validation teams were authorized by Head- QA or Designee.
- All the personnel involved in the manufacturing and packing of validation batches were well trained.
- Sampling and testing of validation samples were done as per their job responsibilities and Authorized by Head-QA or Designee.

9.0 VALIDATION CRITERIA

- Product were manufactured and packed as per as per BMR and BPR.
- Three batches considered for Validation study.
- Current version of standard operating procedure to be followed.
- All activities carried out as per protocol.
- Record the observations stage wise in the respective batch manufacturing and packing record
- Record the yield after Blending, Compression, Coating and packing in manufacturing and packing record.
- The data generated during manufacturing and captured in the executed batch manufacturing records is compiled and verified against the acceptance criteria defined in the process Validation protocol.
- Three optimum batches subjected to stability study
- Validation report prepared based on the analytical reports / observation of Validation batch.

10.0 CRITERIA FOR REVALIDATION

- The process shall be revalidated whenever there is change in;
- Change in site of manufacturing
- Change in batch size.
- Change in Equipment
- Manufacturing process change.
- Manufacturing Formula (Change in composition or components)
- Change in API vendor or critical excipient.
- Change in specification of input material.
- Abnormal trend in quality parameters identified in APQR.
- Trend of Out of specification.

11.0 Manufacturing Formula:

Mother Batch Size: 120000 Tablets

Ingredients	Item code	Grade	Category	Qty. / Batch in kg	Qty. / Tablet in mg
DRY MIX					
Solifenacin Succinate	A036BP01A	BP	Active Substance	*0.600	5.000
Lactose Monohydrate	E101BP	BP	Diluent	12.300	102.500
Maize Starch	E041BP	BP	Bulking agent	#4.800	40.000
BINDER					
Hypromellose	E033BP	BP	Binder	0.120	1.000
Purified water	NA	--	Solvent	@2.458	Q.S.
LUBRICATION					
Magnesium stearate	E040BP	BP	Lubricant	0.180	1.50
Core tablet weight					150.00
Film Coating					
Readycoat Iron Oxide Yellow 36FO193220	E015IH	IH	Film former	**0.468	3.000
Purified water	NA	--	Solvent	@4.212	Q.S.
Total Tablet weight					153.00

Note*: Actual quantity of Solifenacin Succinate to be calculated on the basis of Assay (dried basis) and % LOD.

Compensation will be with maize starch in actual batch.

**30% extra qty. to avoid the loss during coating

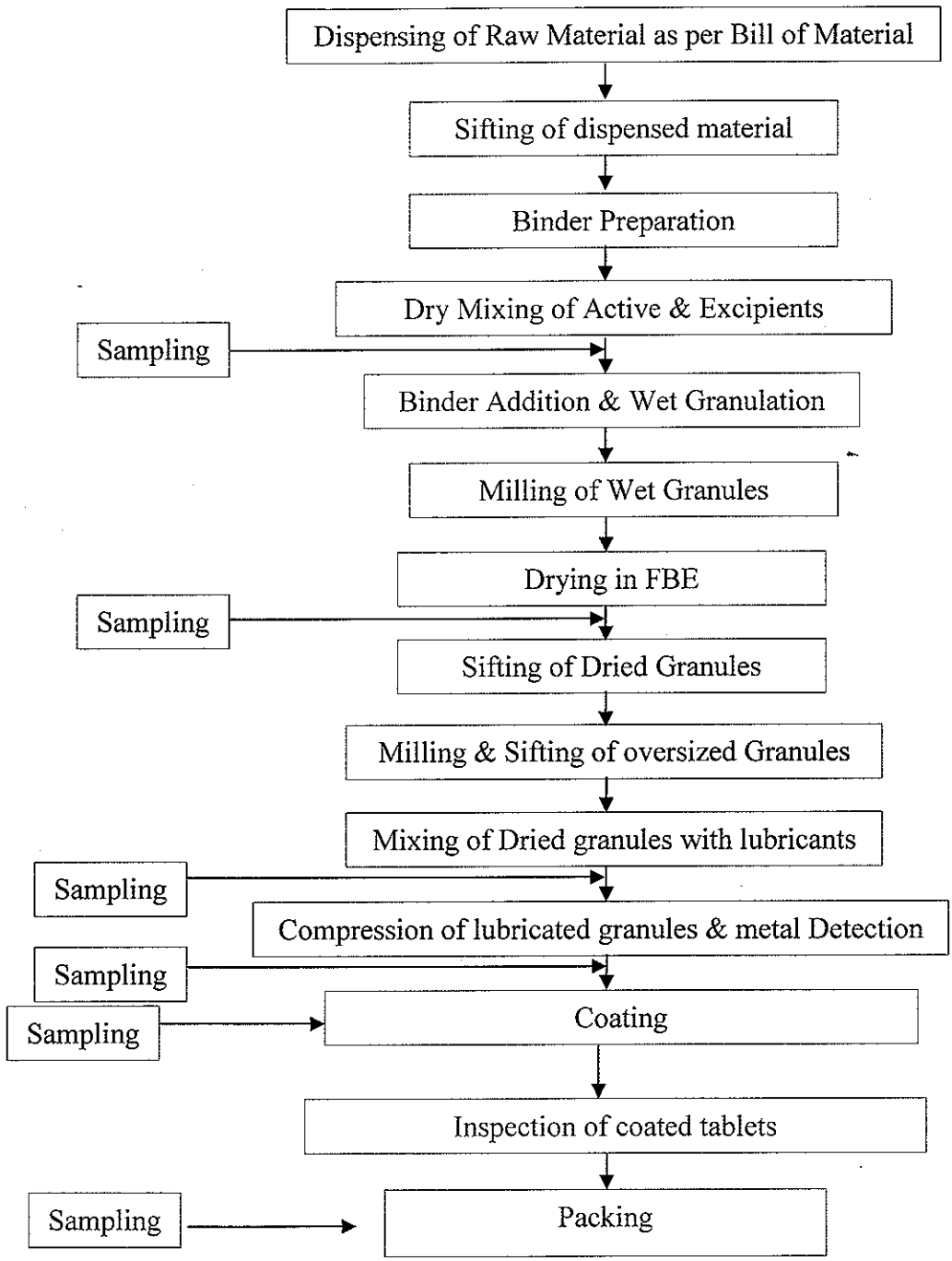
@Quantity not remains in final product.

12.0 REFERENCES DOCUMENTS:

Master Formula Record	MFR/RD/T-05/00
Master packing Record	MPR/RD/T-05/00
Batch Manufacturing Record	BMR/T-05/00
Batch Packing Record	BPR/T-05/00

Note: All current version of relevant documents referred.

13.0 PROCESS FLOW CHART:



14.0 Description of equipment, facility and utilities involved in manufacturing process:

Equipment used in the manufacturing process of Solifenacin Succinate Tablets 5 mg under process validation study was found to be qualified prior to initiation of validation activity. Find below the details.

Sr. No.	Name of the equipment	Equipment ID
Warehouse		
01	OFCB	EQ/WRH/OFCB/004
02	Weighing Balance	INQ/WRH/WEB/003
03	Weighing Balance	INQ/WRH/WEB/004
Production		
01	Double Deck Vibratory Sifter (30 inch)	EQ/OSD/VSF/002
02	High Shear Mixer Granulator - 100L	EQ/OSD/HSMG/001
03	Pillar Type Bin Blender - 300 L	EQ/OSD/PBB/001
04	Binder Preparation Vessel 60 L	EQ/OSD/BPV/001
05	Fluid Bed Equipment (FBE 125 C)	EQ/OSD/FBE/001
06	Tap Cone Mill -100	EQ/OSD/TCM/001
07	Weighing Balance	INQ/OSD/WEB/014
08	Weighing Balance	INQ/OSD/WEB/013
09	27 Station 'D' Tooling Double Rotary Compression Machine	EQ/OSD/COM/001
10	Combo metal detector and de-duster	EQ/OSD/DMD/001 EQ/OSD/DMD/002
11	Weighing Balance	INQ/OSD/WEB/003
12	Capsule and tablet inspection belt	EQ/OSD/CTIC/001
13	Blister Packing Machine	EQ/OSD/RBPM/001
14	Weighing Balance	INQ/OSD/WEB/005
IPQA		
01	Weighing Balance	INQ/IPQA/WEB/001
02	Vernier Caliper	INQ/IPQA/VC/001
03	Disntegration Test Apparatus	INQ/IPQA/DT/001
04	Leak Test Apparatus	INQ/IPQA/LT/001
05	Hardness Tester	INQ/IPQA/THT/001
06	Halogen Moisture Analyzer	INQ/IPQA/HMA/001

Sr. No.	Name of the equipment	Equipment ID
07	Friability Test apparatus	INQ/IPQA/FT/001
Quality Control		
01	Analytical Weighing Balance	INQ/QCD/WEB/002, INQ/QCD/WEB/003
02	HPLC	INQ/QCD/HPLC/001,INQ/QCD/HPLC/002 INQ/QCD/HPLC/006.
03	UV Visible Spectrophotometer	INQ/QCD/UV/001
04	IR	INQ/QCD/IR/001
05	Vernier Caliper	INQ/QCD/VC/002
06	Disintegration test Apparatus	INQ/QCD/DT/001
07	Friability Test apparatus	INQ/QCD/FT/001
08	Hardness Tester	INQ/QCD/THT/001
09	pH meter	INQ/QCD/PH/001
10	Tap Density Apparatus	INQ/QCD/TAP/001
11	Dissolution test apparats	INQ/QCD/DIS/001, INQ/QCD/DIS/002
12	Autotitrator	INQ/QCD/AT/001
13	UV Cabinet	INQ/QCD/UVC/001
14	Stability Chember	EQ/QCD/STB/001 to EQ/QCD/STB/003
Microbiology		
01	Weighing Balance	INQ/QCM/WEB/001, INQ/QCM/WEB/002
02	HLAF	EQ/QCM/HLAF/001
03	Vertical Autoclave	EQ/VA/001
04	Biosafety Cabinet	EQ/QCM/BSC/001

Note: All the manufacturing equipment and instrument qualification status is reviewed and found within validity period.

15.0 List of Approved Vendors:

Sr. No	Name of Raw Material & Grade	Approved Vendor Name & Address
Raw Material (API and Excipients)		
01	SOLIFENACIN SUCCINATE BP	**Amoli Organics Pvt.Ltd.** 407, Dalmal House, Jamnalal Bajaj Road, Nariman Point, Nariman Point, Mumbai, Maharashtra 400021
02	LACTOSE MONOHYDRATE BP	**Prachin Chemicals** 1307/6-7, Phase IV, G.I.D.C., Naroda, Ahmedabad-382330, Gujarat, India.
03	MAIZE STARCH BP	**Shivangan Food and Pharma Products Pvt. Ltd.** Survey No. 226/1, Rawal Industrial Estate, Shahada Road, Dadanagar,Dondaicha-425408 Dist: Dhule (M.S.)
04	HYPROMELLOSE BP	**LOTTE Fine Chemical co., Ltd.** 47, namdong-daero 79beon-gil namdong-gu, Incheon 21700, Republic of Korea
05	MAGNESIUM STEARATE BP	**Nitika Pharmaceutical Specialities Pvt. Ltd.** Block No.03 of Plot No. 1296 1297/08, Ward No.03, Mouza-Waddhamana, Tah-Hingna, Amravati Road, Nagpur-440023.
06	READYCOAT IRON OXIDE YELLOW 36FO193220	**Dhara Life Science Pvt. Ltd. (Unit-I)** Block No.1273, At. Moti Bhoyan, Vadsar-Kalol Road, Ta. Kalol, Dist: Gandhinagar-382721, Gujarat, India.

Sr. No	Packing Material	Approved Vendor Name & Address
01	AF Solifenacin T 5 mg 224 MM	**Shah's Super Seals India Pvt.Ltd.** W-B-24 M.I.D.C., Ambad, Near Nashik Merchant Bank, Nashik-422 010
02	BASE FOIL 226 MM	**Indian Cork Mills Pvt. Ltd.** S.No.57/3(1), Bhenslore Dunetha, Nani Daman 396210
03	Mono Carton (105x75x13mm)	**H.K.Printers Pvt Ltd** H.K House ,inside Neelkanth near sai baba temple ,satavali vasai,Palghar, Mumbai
04	Pack Insert Solifenacin T 5 mg	**Kirti Printers** Near Akola Urban Bank, old cotton market, Akola-444001

16.0 PROCESS VALIDATION METHODOLOGY

- During the course of Validation the documentation systems, laboratory controls, raw materials, final product of Solifenacin Succinate Tablets 5 mg evaluated.
- The Process Validation Methodology consist of three basic parts: process parameters monitoring, the routine in process and final product release testing & additional sampling and testing.
- The validation exercise for three consecutive batches of production scale done to assess the process consistency and quality data of the product; API of different AR No. is used in first and second batches and combination of both A.R. No is used in third validation batch.
- Operate the manufacturing & packing process as per the SOP.

17.0 Brief Description of Manufacturing Procedure:

A. Dispensing of raw materials:

B. No.: LT0101001 Date of material issued: 01/03/2021

(Batch size: 1,20,000 Tablets)

Controlled Variables	Acceptance Criteria	Batch No.
		LT0101001
Area Temperature (°C)	**22 ± 3**	Complies
Area Humidity (%)	**50 ± 10**	Complies

Material Code	Ingredients & Grade	Quantity in Kg	Actual quantity dispensed (Kg)	A.R. No.
DRY MIXING				
A036BP01A	SOLIFENACIN SUCCINATE BP	*0.600	0.602	L21RM00085
E101BP	LACTOSE MONOHYDRATE BP	12.300	12.300	L21RM00091
E041BP	MAIZE STARCH BP	#4.800	#4.798	L21RM00095
BINDER MATERIAL				
E033BP	HYPROMELLOSE BP	0.120	0.120	L21RM00098
LUBRICATION				
E040BP	MAGNESIUM STEARATAE BP	0.180	0.180	L21RM00104
FILM COATING				
E015IH	READYCOAT IRON OXIDE YELLOW 36FO193220 IH	**0.468	0.468	L21RM00107

Note: *Actual quantity of Solifenacin Succinate to be calculated on the basis of Assay (dried basis) and % LOD.

Compensation will be with corn starch in actual batch.

**30% extra qty. to avoid the loss during coating

B. No.: LT0101002 Date of material issued: 02/03/2021

(Batch size: 1,20,000 Tablets)

Controlled Variables	Acceptance Criteria	Batch No.
		LT0101002
Area Temperature (°C)	22 ± 3	Complies
Area Humidity (%)	50 ± 10	Complies

Material Code	Ingredients & Grade	Quantity in Kg	Actual quantity dispensed (Kg)	A.R. No.
DRY MIXING				
A036BP01A	SOLIFENACIN SUCCINATE BP	*0.600	0.601	L21RM00086
E101BP	LACTOSE MONOHYDRATE BP	12.300	12.300	L21RM00091
E041BP	MAIZE STARCH BP	#4.800	#4.799	L21RM00095
BINDER MATERIAL				
E033BP	HYPROMELLOSE BP	0.120	0.120	L21RM00098
LUBRICATION				
E040BP	MAGNESIUM STEARATAE BP	0.180	0.180	L21RM00104
FILM COATING				
E015IH	READYCOAT IRON OXIDE YELLOW 36FO193220 IH	**0.468	0.468	L21RM00107

Note: *Actual quantity of Solifenacin Succinate to be calculated on the basis of Assay (dried basis) and % LOD.

Compensation will be with corn starch in actual batch.

**30% extra qty. to avoid the loss during coating

LEBEN LIFE SCIENCES	**Leben Life Sciences Pvt. Ltd.** **Akola - 444104 (MS) INDIA**		
PROCESS VALIDATION REPORT			
Generic Name	Solifenacin Succinate Tablets 5 Mg	**Report No.**	LLS/T/PVR/005/001
Supersedes	Nil	**Page No.**	15 of 56

B. No.: LT0101003 Date of material issued: 03/03/2021

(Batch size: 1,20,000 Tablets)

Controlled Variables	**Acceptance Criteria**	**Batch No.**
		LT0101003
Area Temperature (°C)	22 ± 3	Complies
Area Humidity (%)	50 ± 10	Complies

Material Code	**Ingredients & Grade**	**Quantity in Kg**	**Actual quantity dispensed (Kg)**	**A.R. No.**
DRY MIXING				
A036BP01A	SOLIFENACIN SUCCINATE BP	*0.600	0.301 0.301	L21RM00085 L21RM00086
E101BP	LACTOSE MONOHYDRATE BP	12.300	12.300	L21RM00091
E041BP	MAIZE STARCH BP	#4.800	#4.798	L21RM00095
BINDER MATERIAL				
E033BP	HYPROMELLOSE BP	0.120	0.120	L21RM00098
LUBRICATION				
E040BP	MAGNESIUM STEARATAE BP	0.180	0.180	L21RM00104
FILM COATING				
E015IH	READYCOAT IRON OXIDE YELLOW 36FO193220 IH	**0.468	0.468	L21RM00107

Note: *Actual quantity of Solifenacin Succinate to be calculated on the basis of Assay (dried basis) and % LOD.

Compensation will be with corn starch in actual batch.

**30% extra qty. to avoid the loss during coating

B. Granulation:

Critical process Parameters:

Sifting of API and Excipients:

Controlled Variables		Acceptance Criteria	Batch No.		
			LT0101001	**LT0101002**	**LT0101003**
Area Temperature (°C)		22 ± 3	Complies	Complies	Complies
Area Humidity (%)		50 ± 10	Complies	Complies	Complies
Sieve Used		40# & 60#	40# & 60#	40# & 60#	40# & 60#
Sieve Integrity	**Before**	Complies/ Not Complies	Complies	Complies	Complies
	After		Complies	Complies	Complies

Dry mixing:

Controlled Variables	Acceptance Criteria	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Area Temperature (°C)	22 ± 3	Complies	Complies	Complies
Area Humidity (%)	50 ± 10	Complies	Complies	Complies
Dry Mixing time (minutes)	10	10	10	10
Impeller Speed	Slow (55 RPM)	55 RPM	55 RPM	55 RPM
Chopper Speed	Off	Off	Off	Off

Description:

Batch No.	Description
LT0101001	White to off white powder
LT0101002	White to off white powder
LT0101003	White to off white powder

Assay of Dry Mixing:

Sampling Points	% Obtained		
	Batch No. LT0101001	**Batch No. LT0101002**	**Batch No. LT0101003**
	Solifenacin Succinate	**Solifenacin Succinate**	**Solifenacin Succinate**
Top	101.6	101.0	101.5
Middle	102.5	101.9	101.4
Bottom	102.9	100.3	102.1
Composite	101.5	101.1	101.8

Remark: Assay result of Solifenacin Succinate is found within the limits. Hence, based on above results it is concluded that dry mixing time 10 minutes is sufficient to produce uniform mixture.

Binder Preparation:

Controlled Variables	Acceptance Criteria	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Area Temperature (°C)	22 ± 3	Complies	Complies	Complies
Area Humidity (%)	50 ± 10	Complies	Complies	Complies
Total Quantity of Purified Water	2.458 Kg	2.458 Kg	2.458 Kg	2.458 Kg
Clarity of Solution	Should be Uniform	Uniform	Uniform	Uniform
Purified Water A.R No.		L21PW00120	L21PW00126	L21PW00132

Wet Mixing & Wet Mass Discharging Activity:

Controlled Variables	Acceptance Criteria	Batch No.		
		LT0101001	LT0101002	LT0101003
Area Temperature (°C)	22 ± 3	Complies	Complies	Complies
Area Humidity (%)	50 ± 10	Complies	Complies	Complies
Mixing Time Including Binder Addition Time	2 Minutes	2 Minutes	2 Minutes	2 Minutes
Impeller Speed during addition of Binder Solution	55 RPM	55 RPM	55 RPM	55 RPM
Chopper Speed during addition of Binder Solution	OFF	OFF	OFF	OFF
Kneading Time	30 second	1 Minute	1 Minute	1 Minute
Impeller Speed	To be recorded	196 RPM	196 RPM	196 RPM
Chopper Speed	To be recorded	1850 RPM	1850 RPM	1850 RPM
Amperage	To be recorded	18.85A	19.31A	20.12A

Drying:

Controlled Variables		Acceptance Criteria	Batch No.		
			LT0101001	LT0101002	LT0101003
Area Temperature (°C)		22 ± 3	Complies	Complies	Complies
Area Humidity (%)		50 ± 10	Complies	Complies	Complies
Drying Time (Min) Set Point		To be recorded	30 min	30 min	20 min
Inlet temperature		50-70°C	60	60	60
Outlet temperature		35-50 °C	36	38	37
LOD	Production	NMT 2%	1.018	1.493	1.218
	QA		0.978	1.317	1.177

Sizing of Dried Granules:

Controlled Variables		Acceptance Criteria	Batch No.		
			LT0101001	LT0101002	LT0101003
Area Temperature (°C)		22 ± 3	Complies	Complies	Complies
Area Humidity (%)		50 ± 10	Complies	Complies	Complies
Screen Used		2.0 mm	2.0 mm	2.0 mm	2.0 mm
Screen Integrity	Before	Complies/Not Complies	Complies	Complies	Complies
	After		Complies	Complies	Complies
Sieve Used		20#	20#	20#	20#
Sieve Integrity	Before	Complies/Not Complies	Complies	Complies	Complies
	After		Complies	Complies	Complies
Milling Speed		750 RPM to 1500 RPM	1104 RPM	1197 RPM	1096 RPM

Lubrication:

Stage	Controlled Variables		Acceptance Criteria	Batch No.		
				LT0101001	**LT0101002**	**LT0101003**
Lubrication	**Area Temperature (°C)**		22 ± 3	Complies	Complies	Complies
	Area Humidity (%)		50 ± 10	Complies	Complies	Complies
	Sieve Used		20#	20#	20#	20#
	Sieve Integrity	**Before**	Complies/Not Complies	Complies	Complies	Complies
		After		Complies	Complies	Complies
	Blending Time		5 Minutes	5 Minutes	5 Minutes	5 Minutes
	Blender Speed		08 RPM	08 RPM	08 RPM	08 RPM
	LOD		NMT 2.0%	1.428%	1.312%	1.416%

QC Result of Blend:

Appearance:

Batch No.	Appearance of lubricated granules
LT0101001	White to off white granules.
LT0101002	White to off white granules.
LT0101003	White to off white granules.

Bulk Density and Tapped density:

LT0101001		LT0101002		LT0101003	
BD gm/ml	**TD gm/ml**	**BD gm/ml**	**TD gm/ml**	**BD gm/ml**	**TD gm/ml**
0.528	0.606	0.577	0.732	0.504	0.715

Water content:

LT0101001	LT0101002	LT0101003
Water content (NMT 5.0%)	**Water content (NMT 5.0%)**	**Water content (NMT 5.0%)**
1.5275%	1.4650%	1.3599%

Uniformity of Mixing by Assay:

Sampling points	% Obtained		
	LT0101001	**LT0101002**	**LT0101003**
	Solifenacin Succinate	**Solifenacin Succinate**	**Solifenacin Succinate**
Top T1	103.0	99.4	99.9
Top T2	102.5	96.7	100.2
Top T3	101.1	98.0	99.9
Top T4	103.3	95.8	99.0
Middle M1	103.0	97.5	101.1
Middle M2	102.2	96.3	100.6
Bottom B1	102.1	97.5	100.4
Bottom B2	101.9	98.1	101.0
Bottom B3	102.9	98.6	100.2
Bottom B4	102.5	99.2	100.5
Composite	101.5	95.3	100.3
Mean	102.36	97.49	100.28
SD	0.68	1.35	0.57
%RSD	0.66	1.38	0.57

Remark: Assay result of Solifenacin Succinate is found within the limits and RSD value is less than 5%. Hence, based on above results it is concluded that mixing time 20 minutes is sufficient to produce uniform mixture.

Weighing of Lubricated granules:

Theoretical weight of Lubricated granules: 18.000 kg

Batch No.	Total weight of lubricated granules (Kg)	% Yield
LT0101001	17.9046	99.47
LT0101002	17.8236	99.02
LT0101003	17.8074	98.93

Remark: Based upon the % observed yield it clearly indicates that the process gives consistent yield.

C. Compression:

Stage	Controlled Variables	Acceptance Criteria	Batch No.		
			LT0101001	**LT0101002**	**LT0101003**
Compression	**Area Temperature (°C)**	22 ± 3	Complies	Complies	Complies
	Area Humidity (%)	50 ± 10	Complies	Complies	Complies
	Punch Size	7.6 mm	7.6 mm	7.6 mm	7.6 mm
	Punch and Die Shape	Round shaped	Round shaped	Round shaped	Round shaped
	Upper Punch Description	Round shaped plain	Round shaped plain	Round shaped plain	Round shaped plain
	Lower Punch Description	Round shaped plain	Round shaped plain	Round shaped plain	Round shaped plain
	Type of Tooling	"B" type	"B" type	"B" type	"B" type
	No. of Punches 27 station	27 Nos	27 Nos	27 Nos	27 Nos
	Speed Range	16-42 RPM	16-20 RPM	16-20 RPM	16-20 RPM

➢ **Process results and observation at compression stage:**

RHS

1. **(Speed Challenge Test) Low Speed at 16 RPM**

Pressure: 4.6 Ton **Thickness Adjustment:** Front Side:12 , Rear side: 12

Feeder speed: - 30 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 139 mg	Min: 140 mg	Min:141 mg
		Max: 142 mg	Max: 143 mg	Max: 144 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.7 mg	150.8 mg	150.8 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.42 mm	Min: 3.44 mm	Min: 3.47 mm
		Max: 3.51 mm	Max: 3.53 mm	Max: 3.55 mm
Hardness	NLT 3.0 kp	Min: 9.682 Kp	Min: 12.655 Kp	Min: 10.873 Kp
		Max: 15.042 Kp	Max: 18.042 Kp	Max: 16.294 Kp
Disintegration Time	NMT 15 min.	01 min. 58 sec	01 min. 40 sec	02 min. 32 sec
Friability	NMT 1.0% w/w	0.30%	0.16%	0.13%
Conclusion: Process parameters found complying with the predefined specification.				

2. (Speed Challenge Test) Low Speed at 16 RPM

LHS

Pressure: 4.6 Ton **Thickness Adjustment:** Front Side:12 , Rear side: 12

Feeder speed: - 30 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 139 mg	Min: 141 mg	Min: 142 mg
		Max: 145 mg	Max: 146 mg	Max: 143 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	149.7 mg	150.8 mg	150.7 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.46 mm	Min: 3.41 mm	Min: 3.39 mm
		Max: 3.53 mm	Max: 3.51 mm	Max: 3.46 mm
Hardness	NLT 3.0 kp	Min: 10.459 Kp	Min: 13.358 Kp	Min: 12.599 Kp
		Max: 15.118 Kp	Max: 17.292 Kp	Max: 16.241 Kp
Disintegration Time	NMT 15 min.	02 min. 18 sec	02 min. 03 sec	02 min. 34 sec
Friability	NMT 1.0% w/w	0.13%	0.12%	0.11%
Conclusion: Process parameters found complying with the predefined specification.				

3. (Speed Challenge Test) High Speed at 20 RPM

RHS

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 34 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 139 mg	Min: 138 mg	Min: 138 mg
		Max: 142 mg	Max: 143 mg	Max: 145 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.2 mg	150.4 mg	149.6 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.39 mm	Min: 3.44 mm	Min: 3.48 mm
		Max: 3.48 mm	Max: 3.52 mm	Max: 3.55 mm
Hardness	NLT 3.0 kp	Min: 9.304 Kp	Min:10.422 Kp	Min: 9.354 Kp
		Max: 13.909 Kp	Max: 15.698 Kp	Max: 14.203 Kp
Disintegration Time	NMT 15 min.	02 min. 20 sec	02 min. 23 sec	02 min. 21 sec
Friability	NMT 1.0% w/w	0.19%	0.13%	0.15%
Conclusion: Process parameters found complying with the predefined specification.				

4. (Speed Challenge Test) High Speed at 20 RPM

LHS

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 34 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 139 mg	Min: 143 mg	Min: 142 mg
		Max: 148 mg	Max: 147 mg	Max: 146 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	148.9 mg	150.8 mg	150.5 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.39 mm	Min: 3.45 mm	Min: 3.47 mm
		Max: 3.53 mm	Max: 3.51 mm	Max: 3.54 mm
Hardness	NLT 3.0 kp	Min: 10.751 Kp	Min: 11.470 Kp	Min: 12.223 Kp
		Max: 14.541 Kp	Max: 15.478 Kp	Max: 15.603 Kp
Disintegration Time	NMT 15 min.	02 min. 09 sec	02 min. 11 sec	02 min. 17 sec
Friability	NMT 1.0% w/w	0.12%	0.16%	0.14%
Conclusion: Process parameters found complying with the predefined specification.				

Remarks:

Based on above compression speed challenge tests, below parameters are established.

Minimum compression speed at **16 RPM**

Maximum compression speed at **20 RPM**

Optimum compression speed at **18 RPM** shall be considered for further challenge test.

5. (Speed Challenge Test) Optimum Speed at 18 RPM

RHS

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 140 mg	Min: 141 mg	Min: 142 mg
		Max: 145 mg	Max: 148 mg	Max: 147 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.7 mg	150.8 mg	150.2 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.46 mm	Min: 3.43 mm	Min: 3.47 mm
		Max: 3.52 mm	Max: 3.53 mm	Max: 3.54 mm
Hardness	NLT 3.0 kp	Min: 7.583 Kp	Min: 11.516 Kp	Min: 11.549 Kp
		Max: 13.545 Kp	Max: 15.087 Kp	Max: 15.278 Kp
Disintegration Time	NMT 15 min.	02 min. 11 sec	02 min. 07 sec	02 min. 22 sec
Friability	NMT 1.0% w/w	0.17%	0.11%	0.10%
Conclusion: Process parameters found complying with the predefined specification.				

6. (Speed Challenge Test) Optimum Speed at 18 RPM :

LHS

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 142 mg	Min: 141 mg	Min: 140 mg
		Max: 148 mg	Max: 149 mg	Max: 145 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.2 mg	150.4 mg	150.7 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.41 mm	Min: 3.42 mm	Min: 3.45 mm
		Max: 3.48 mm	Max: 3.51 mm	Max: 3.53 mm
Hardness	NLT 3.0 kp	Min: 10.622 Kp	Min: 11.776 Kp	Min: 11.811 Kp
		Max: 14.779 Kp	Max: 15.643 Kp	Max: 15.647 Kp
Disintegration Time	NMT 15 min.	02 min. 20 sec	01 min. 54 sec	02 min. 24 sec
Friability	NMT 1.0% w/w	0.24%	0.15%	0.12%
Conclusion: Process parameters found complying with the predefined specification.				

7. Low Hardness (RHS):

Optimum Speed : 18 RPM

Pressure: 4.5 Ton **Thickness Adjustment:** Front Side: 17, Rear side: 17

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 139 mg	Min: 141 mg	Min: 142 mg
		Max: 148 mg	Max: 146 mg	Max: 149 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.4 mg	150.6 mg	148.6 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.42 mm	Min: 3.46 mm	Min: 3.44 mm
		Max: 3.51 mm	Max: 3.52 mm	Max: 3.54 mm
Hardness	NLT 3.0 kp	Min: 8.120 Kp	Min: 8.514 Kp	Min: 6.529 Kp
		Max: 12.269 Kp	Max: 12.312 Kp	Max: 11.706 Kp
Disintegration Time	NMT 15 min.	01 min. 27 sec	01 min. 25 sec	02 min. 58 sec
Friability	NMT 1.0% w/w	0.39%	0.21%	0.29%
Conclusion: Process parameters found complying with the predefined specification.				

8. Low Hardness (LHS):

Optimum Speed: 18 RPM

Pressure: 4.5 Ton **Thickness Adjustment:** Front Side: 17, Rear side: 17

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 139 mg	Min: 141 mg	Min:145 mg
		Max: 145 mg	Max: 145 mg	Max:149 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.3 mg	149.8 mg	150.0 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.40 mm	Min: 3.43 mm	Min: 3.45 mm
		Max: 3.52 mm	Max: 3.52 mm	Max: 3.54 mm
Hardness	NLT 3.0 kp	Min: 7.034 Kp	Min: 9.552 Kp	Min: 6.780 Kp
		Max: 11.948 Kp	Max: 11.313 Kp	Max: 10.188 Kp
Disintegration Time	NMT 15 min.	01 min.25 sec	01 min. 30 sec	01 min. 47 sec
Friability	NMT 1.0% w/w	0.31%	0.18%	0.31%
Conclusion: Process parameters found complying with the predefined specification.				

9. **High Hardness (RHS):**

Optimum Speed : 18 RPM

Pressure: 6 Ton **Thickness Adjustment:** Front Side: 8, Rear side: 8

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 139 mg	Min: 143 mg	Min: 148 mg
		Max: 144 mg	Max: 156 mg	Max: 151 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.2 mg	150.4 mg	150.8 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.34 mm	Min: 3.36 mm	Min: 3.38 mm
		Max: 3.40 mm	Max: 3.40 mm	Max: 3.43 mm
Hardness	NLT 3.0 kp	Min: 12.790 Kp	Min: 13.249 Kp	Min: 12.538 Kp
		Max: 17.211 Kp	Max: 19.313 Kp	Max: 18.286 Kp
Disintegration Time	NMT 15 min.	02 min. 56 sec	03 min. 27 sec	03 min. 47 sec
Friability	NMT 1.0% w/w	0.12%	0.11%	0.16%
Conclusion: Process parameters found complying with the predefined specification.				

10. High Hardness (LHS):

Optimum Speed: 18 RPM

Pressure: 6 Ton **Thickness Adjustment:** Front Side: 8, Rear side: 8

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 142 mg	Min: 140 mg	Min: 137 mg
		Max: 149 mg	Max: 147 mg	Max: 148 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.9 mg	149.2 mg	150.1 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.36 mm	Min: 3.37 mm	Min: 3.36 mm
		Max: 3.42 mm	Max: 3.42 mm	Max: 3.43 mm
Hardness	NLT 3.0 kp	Min: 11.883 Kp	Min: 12.946 Kp	Min: 12.936 Kp
		Max: 17.866 Kp	Max: 17.555 Kp	Max: 19.015 Kp
Disintegration Time	NMT 15 min.	02 min. 11 sec	02 min. 14 sec	02 min. 56 sec
Friability	NMT 1.0% w/w	0.11%	0.13%	0.14%
Conclusion: Process parameters found complying with the predefined specification.				

Remarks:

Based on above hardness challenge tests, below parameters are established.

Minimum Hardness: 6.529 Kp

Maximum Hardness: 19.313 Kp

Optimum Hardness: 12.921 Kp

11. Initial RHS:

Optimum Speed: 18 RPM

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 140 mg	Min: 144 mg	Min: 143 mg
		Max: 152 mg	Max: 153 mg	Max: 155 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.7 mg	150.5 mg	150.8 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.46 mm	Min: 3.40 mm	Min: 3.47 mm
		Max: 3.55 mm	Max: 3.54 mm	Max: 3.56 mm
Hardness	NLT 3.0 kp	Min: 9.726 Kp	Min: 9.945 Kp	Min: 10.494 Kp
		Max: 14.792 Kp	Max: 15.697 Kp	Max: 14.436 Kp
Disintegration Time	NMT 15 min.	02 min. 14 sec	02 min. 22 sec	02 min. 29 sec
Friability	NMT 1.0% w/w	0.14%	0.13%	0.14%
Conclusion: Process parameters found complying with the predefined specification.				

12. Initial LHS:

Optimum Speed : 18 RPM

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 143 mg	Min: 144 mg	Min: 145 mg
		Max: 150 mg	Max: 154 mg	Max: 156 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	149.4 mg	150.1 mg	150.7 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.49 mm	Min: 3.48 mm	Min: 3.49 mm
		Max: 3.56 mm	Max: 3.57 mm	Max: 3.54 mm
Hardness	NLT 3.0 kp	Min: 11.218 Kp	Min: 12.216 Kp	Min: 12.410 Kp
		Max: 15.511 Kp	Max: 14.998 Kp	Max: 14.386 Kp
Disintegration Time	NMT 15 min.	02 min. 03 sec	02 min. 14 sec	02 min. 12 sec
Friability	NMT 1.0% w/w	0.13%	0.16%	0.14%
Conclusion: Process parameters found complying with the predefined specification.				

13. **Middle RHS:**

Optimum Speed : 18 RPM

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 139 mg	Min: 144 mg	Min: 140 mg
		Max: 148 mg	Max: 149 mg	Max: 152 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.4 mg	150.8 mg	150.1 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.44 mm	Min: 3.47 mm	Min: 3.43 mm
		Max: 3.52 mm	Max: 3.54 mm	Max: 3.54 mm
Hardness	NLT 3.0 kp	Min: 9.360 Kp	Min: 10.341 Kp	Min: 10.482 Kp
		Max: 14.795 Kp	Max: 16.244 Kp	Max: 17.399 Kp
Disintegration Time	NMT 15 min.	02 min.30 sec	02 min. 30 sec	02 min. 42 sec
Friability	NMT 1.0% w/w	0.17%	0.15%	0.12%
Conclusion: Process parameters found complying with the predefined specification.				

14. Middle LHS:

Optimum Speed: 18 RPM

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 143 mg	Min: 142 mg	Min: 141 mg
		Max: 146 mg	Max: 145 mg	Max: 150 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.2 mg	150.3 mg	150.4 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.39 mm	Min: 3.38 mm	Min: 3.39 mm
		Max: 3.47 mm	Max: 3.45 mm	Max: 3.51 mm
Hardness	NLT 3.0 kp	Min: 9.255 Kp	Min: 10.292 Kp	Min: 12.866 Kp
		Max: 15.451 Kp	Max: 17.919 Kp	Max: 14.485 Kp
Disintegration Time	NMT 15 min.	02 min. 17 sec	02 min. 11sec	02 min. 39 sec
Friability	NMT 1.0% w/w	0.14%	0.10%	0.15%
Conclusion: Process parameters found complying with the predefined specification.				

15. End RHS:

Optimum Speed : 18 RPM

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 143 mg	Min: 144 mg	Min: 143 mg
		Max: 148 mg	Max: 148 mg	Max: 149 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.6 mg	150.4 mg	150.6 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.48 mm	Min: 3.47 mm	Min: 3.46 mm
		Max: 3.53 mm	Max: 3.51 mm	Max: 3.54 mm
Hardness	NLT 3.0 kp	Min: 11.125 Kp	Min: 11.139 Kp	Min: 12.044 Kp
		Max:16.141 Kp	Max: 16.486 Kp	Max: 16.885 Kp
Disintegration Time	NMT 15 min.	02 min. 30 sec	01 min. 45 sec	02 min. 34 sec
Friability	NMT 1.0% w/w	0.13%	0.15%	0.11%
Conclusion: Process parameters found complying with the predefined specification.				

16. End LHS:

Optimum Speed: 18 **RPM**

Pressure: 5 Ton **Thickness Adjustment:** Front Side: 12, Rear side: 12

Feeder speed: - 32 RPM

Test	Specifications	Batch No.		
		LT0101001	**LT0101002**	**LT0101003**
Appearance	White to off white, round shaped tablet plain on both sides	Complies	Complies	Complies
Uniformity of Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	Min: 143 mg	Min: 145 mg	Min: 147 mg
		Max: 147 mg	Max: 154 mg	Max: 158 mg
Average weight of 20 tablet	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)	150.8 mg	150.5 mg	149.8 mg
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)	Min: 3.46 mm	Min: 3.40 mm	Min: 3.48 mm
		Max: 3.56 mm	Max: 3.53 mm	Max: 3.54 mm
Hardness	NLT 3.0 kp	Min: 9.485 Kp	Min: 11.121 Kp	Min: 10.855 Kp
		Max: 15.094 Kp	Max: 16.267 Kp	Max: 15.771 Kp
Disintegration Time	NMT 15 min.	02 min. 27 sec	02 min. 20 sec	02 min. 27 sec
Friability	NMT 1.0% w/w	0.11%	0.11%	0.13%
Conclusion: Process parameters found complying with the predefined specification.				

QC Speed Challenge Results:

Batch No.	Speed		
	Minimum	**Maximum**	**Optimum**
	Assay (95.0 to 105.0%)	**Assay (95.0 to 105.0%)**	**Assay (95.0 to 105.0%)**
	Solifenacin Succinate	**Solifenacin Succinate**	**Solifenacin Succinate**
LT0101001	99.7	97.8	99.4
LT0101002	100.1	100.1	100.0
LT0101003	101.1	99.9	100.3

Batch No.	Speed		
	Minimum	**Optimum**	**Maximum**
	Description	**Description**	**Description**
LT0101001	Complies	Complies	Complies
LT0101002	Complies	Complies	Complies
LT0101003	Complies	Complies	Complies

QC Hardness Challenge Results:

Batch No	Lower Hardness		Higher Hardness	
	Assay (95-105%) Solifenacin Succinate	Average Weight (mg)	Assay (95-105%) Solifenacin Succinate	Average Weight (mg)
LT0101001	102.6	150.28	101.1	150.07
LT0101002	102.0	150.35	102.0	150.36
LT0101003	98.7	150.09	99.9	150.15

Batch No	Lower Hardness	Higher Hardness
	Description	**Description**
LT0101001	Complies	Complies
LT0101002	Complies	Complies
LT0101003	Complies	Complies

QC Initial, middle and End Results:

Batch No	Initial	Middle	End
	Assay (95.0 to 105.0%)	**Assay (95.0 to 105.0%)**	**Assay (95.0 to 105.0%)**
	Solifenacin Succinate	**Solifenacin Succinate**	**Solifenacin Succinate**
LT0101001	102.4	101.1	100.7
LT0101002	100.3	101.9	99.8
LT0101003	99.5	100.4	99.9

Batch No	Average Weight (mg)			
	Initial	Middle	End	Composite
LT0101001	150.838	150.738	150.953	150.261
LT0101002	150.423	150.254	150.231	150.283
LT0101003	150.581	150.701	150.065	150.631

Batch No	Initial	Middle	End
	Description	Description	Description
LT0101001	Complies	Complies	Complies
LT0101002	Complies	Complies	Complies
LT0101003	Complies	Complies	Complies

Composite Sample:

Tests	Batch No.		
	LT0101001	LT0101002	LT0101003
Description	Complies	Complies	Complies
Water Content	1.3137%	1.6184%	1.5730%
Average Weigh (mg)	150.261	150.28	150.463
Assay	**Solifenacin Succinate**	**Solifenacin Succinate**	**Solifenacin Succinate**
	101.5%	101.6%	99.0%

Batch Yield:

Batch No.	LT0101001	LT0101002	LT0101003
Yield %	98.92%	98.53%	98.09%

Remark: Based upon the % observed yield it clearly indicates that the process gives consistent yield.

D. COATING:-

Controlled Variables	Acceptance Criteria	Batch No.		
		LT0101001	LT0101002	LT0101003
		Date:08/03/2021	Date:09/03/2021	Date:10/03/2021
Area Temperature °C	22 ± 3	Complies	Complies	Complies
Area Humidity %	50 ±10	Complies	Complies	Complies

Batch No.:- LT0101001

- **Coating solution preparation:**

STEP NO.	MANUFACTURING PROCEDURE		OBSERVATION
1	Take the 4.212 kg purified water in solution preparation vessel		OK
2	Add Readycoat Iron Oxide Yellow 36FO193220 in purified water under stirring in solution prepartion vessel till clear solution	**Start Time**	10:25
		End Time	10:55
		Stirrer RPM	NA
3	Check the integrity of S.S. SCREEN before filter operation.	**Integrity**	OK
4	After getting the solution, filter the solution through S.S. SCREEN	**Start Time**	10:56
		End Time	11:06
5	Verify integrity of S.S SCREEN after filter operation	**Integrity**	OK

Batch No.:- LT0101002

➢ **Coating solution preparation:**

STEP NO.	MANUFACTURING PROCEDURE		OBSERVATION
1	Take the 4.212 kg purified water in solution preparation vessel		OK
2	Add Readycoat Iron Oxide Yellow 36FO193220 in purified water under stirring in solution prepartion vessel till clear solution	**Start Time**	10:20
		End Time	10:50
		Stirrer RPM	NA
3	Check the integrity of S.S. SCREEN before filter operation.	**Integrity**	OK
4	After getting the solution, filter the solution through S.S. SCREEN	**Start Time**	10:51
		End Time	11:01
5	Verify integrity of S.S SCREEN after filter operation	**Integrity**	OK

Batch No.:- LT0101003

- **Coating solution preparation:**

STEP NO.	MANUFACTURING PROCEDURE		OBSERVATION
1	Take the 4.212 kg purified water in solution preparation vessel		OK
2	Add Readycoat Iron Oxide Yellow 36FO193220 in purified water under stirring in solution prepartion vessel till clear solution	**Start Time**	10:20
		End Time	10:50
		Stirrer RPM	NA
3	Check the integrity of S.S. SCREEN before filter operation.	**Integrity**	OK
4	After getting the solution, filter the solution through S.S. SCREEN	**Start Time**	10:51
		End Time	11:01
5	Verify integrity of S.S SCREEN after filter operation	**Integrity**	OK

- **Gun Validation:**

Batch No.	Nozzle No.	Peristaltic Pump RPM	Time		Tare wt. (gm)	Gross wt. (gm)	Net wt. (gm)	Spray rate gm / minute	Total Net Weight gm.
			From	To					
LT0101001	**1.**	25	11:16	11:17	1.606	52.165	50.559	50.579	101.158
	2.				1.609	52.208	50.599		
LT0101002	**1.**	25	11:16	11:17	1.608	52.155	50.547	50.571	101.142
	2.				1.610	52.205	50.595		

Batch No.	Nozzle No.	Peristaltic Pump RPM	Time		Tare wt. (gm)	Gross wt. (gm)	Net wt. (gm)	Spray rate gm / minute	Total Net Weight gm.
			From	To					
LT0101003	1.	25	11:11	11:12	1.609	52.210	50.601	50.580	101.161
	2.				1.610	52.170	50.560		

➢ **Coating process parameters:**

Parameters	Standard Setting Parameters of Dosing for Auto coater 36"			
	Batch No.	LT0101001	LT0101002	LT0101003
	Standard Limit	Setting Parameter	Setting Parameter	Setting Parameter
Inlet Air temp	60^0C to 80^0C	78.5	79.9	79.8
Pan RPM	2-8 RPM	4	4	4
Atomizing air pressure kg/cm^2	0.1 to 0.4	0.1	0.1	0.1
Peristaltic pump speed (RPM)	5 to 40 RPM	13	14	14

In process Observation:

Parameters	**Batch No.**		
	LT0101001	**LT0101002**	**LT0101003**
Inlet Air Temperature (°C)	Min: 78.6 Max: 79.5	Min:78.9 Max:79.5	Min:78.3 Max:79.9
Exhaust Air Temperature (°C)	Min: 55.8 Max: 56.3	Min:54.8 Max:55.9	Min:56.8 Max:57.1
PAN RPM	Min: 2 Max: 4	Min: 2 Max: 4	Min: 2 Max: 4
Peristalic Pump(RPM)	Min: 10 Max: 14	Min: 10 Max: 14	Min: 10 Max: 14
Average Coated Weight of 20 tablets	153.9	153.2	153.2
% Weight Gain	1.57	1.61	1.51

➢ **IN PROCESS CHECKS FOR FILMCOATING**

Parameter	**Acceptance criteria/Limits**	**Batch No.**					
		LT0101001		**LT0101002**		**LT0101003**	
Appearance	**Light yellow coloured,round shaped film coated tablet,plain on both sides.**	Complies		Complies		Complies	
Average Weight of 20 tablets	**153.0 mg ± 7.5% (141.52 mg to 164.47 mg)**	153.254		153.436		153.212	
Disintrgra-tion	**NMT 30 min**	03 min 12 sec		03 min 02 sec		03 min 17 sec	
Thickness (5 Tab mm)	**3.28 mm ± 0.3 mm (2.98 mm to 3.58 mm)**	Min:-3.42	Max:-3.54	Min:-3.37	Max:-3.50	Min:-3.41	Max:-3.47

➢ **Process yield (Coating Stage):**

Batch No.	% Batch Yield
LT0101001	97.60 %
LT0101002	97.57 %
LT0101003	97.14 %

Remark: Based upon the % observed yield it clearly indicates that the process gives consistent yield

➢ **Assay of Film coated tablet:**

Test	% Obtained (Limit: 95-105%)		
	LT0101001	**LT0101002**	**LT0101003**
Description	Complies	Complies	Complies
Water Content	1.2612%	1.2705%	1.4182%
API	**Solifenacin Succinate**	**Solifenacin Succinate**	**Solifenacin Succinate**
Assay	99.4%	99.3%	99.9%
Dissolution	99.1%	98.6%	99.0%

E. Packing :

Dispensing of packing material

B. No.: LT010E1001 Date of material issued: 01/03/2021

(Export Stability Batch Size: 3000 Tablets)

Material Code	Material Name	Std. Qty.	Actual quantity dispensed	A.R. No.
PM000088	AF Solifenacin T 5 mg 224 MM	0.120 Kg	12.252 kg	L21PM00034
PM000089	BASE FOIL 226 MM	0.600 Kg	20.140 kg	L21PM00035
PM000090	Mono Carton (105x75x13mm)	300 Nos.	306 Nos	L21PM00036
PM000091	Pack Insert Solifenacin T 5 mg	300 Nos.	306 Nos.	L21PM00031

B. No.: LT010E1002 Date of material issued: 02/03/2021

(Export Stability Batch Size: 3000 Tablets)

Material Code	Material Name	Std. Qty.	Actual quantity dispensed	A.R. No.
PM000088	AF Solifenacin T 5 mg 224 MM	0.120 Kg	12.152 kg	L21PM00034
PM000089	BASE FOIL 226 MM	0.600 Kg	19.140 kg	L21PM00035

PM000090	Mono Carton (105x75x13mm)	300 Nos.	306 Nos	L21PM00036
PM000091	Pack Insert Solifenacin T 5 mg	300 Nos.	306 Nos.	L21PM00031

B. No.: LT010E1003 Date of material issued: 03/03/2021

(Export Stability Batch Size: 6000 Tablets)

Material Code	Material Name	Std. Qty.	Actual quantity dispensed	A.R. No.
PM000088	AF Solifenacin T 5 mg 224 MM	0.120 Kg	12.082 kg	L21PM00034
PM000089	BASE FOIL 226 MM	0.600 Kg	18.240 kg	L21PM00035
PM000090	Mono Carton (105x75x13mm)	300 Nos.	306 Nos	L21PM00036
PM000091	Pack Insert Solifenacin T 5 mg	300 Nos.	306 Nos.	L21PM00031

➢ **Challenge test for Blister Packing Machine**

Challenge test for Blister Packing Machine

- Initial Checks to be performed before start of packing.

➢ **In process observation:**

Challenge Test	**Procedure**	**Acceptance Criteria**	**Observation**		
		Batch No.	**LT010E1001**	**LT010E1002**	**LT010E1003**
Alu. Plain Foil & Base Foil end	End of Alu. Plain Foil and Base Foil	Machine stops at zero position	Complies	Complies	Complies
Alu. Plain Foil & Base Foil splice detection	Splice detector shall detect and reject the blister	The blister with splice should get rejected at rejection station	Complies	Complies	Complies
Leak test	Perform the leak test using leak test apparatus Qty- 02	No blister pocket should be leak	Complies	Complies	Complies

In-process checks for Blister

Test	**Specification**	**LT010E1001**	**LT010E1002**	**LT010E1003**
Sealing Temperature	_____To_____ °C	160 °C	160 °C	160 °C
Cutting	Should be proper (Found OK/Not OK)	Ok	Ok	Ok
Knurling	Should be proper (Found OK/Not OK)	Ok	Ok	Ok
Speed of Machine	____ To_____ Punch /min.	30	30	30
Leak Test	Blister Quantity-02	02	02	02
	Leak test (Pass/Fail)	Pass	Pass	Pass

➢ **Packing Yield Observation:**

(Export Stability Batch size: 3000 Tablets)

Batch No.	Batch No. LT010E1001	Batch No. LT010E1002	Batch No. LT010E1003
Yield	99.8%	99.9%	99.6%

18.0 Finished Product Analysis Report

Sr. No.	Tests	Specifications	Batch No. LT010E1001	Batch No. LT010E1002	Batch No. LT010E1003
1	**Description**	Light yellow coloured,round shaped film coated tablets,plain on both sides. Such 10 tablets packed in ALU/LU Blister.	Light yellow coloured,round shaped film coated tablets,plain on both sides. Such 10 tablets packed in ALU/ALU Blister.	Light yellow coloured,round shaped film coated tablets,plain on both sides. Such 10 tablets packed in ALU/ALU Blister.	Light yellow coloured,round shaped film coated tablets,plain on both sides. Such 10 tablets packed in ALU/ALU Blister.
2	**Identification by IR**	The infrared absorption spectrum of the dried residue is concordant with the reference spectrum of solifenacin WS/RS.	Complies	Complies	Complies
3	**Average weight**	153.00 mg ± 7.5% (141.53 mg to 164.48 mg)	152.342 mg	152.561 mg	153.439 mg
4	**Uniformity of weight**	Not More than 2 tablets deviate from average weight by 7.5% None Tablet deviate from the average weight by more than 15 %	Min=147.33 mg Max=158.41 mg Average=152.342 mg	Min=146.24 mg Max=157.38 mg Average=152.561 mg	Min=147.42 mg Max=158.36 mg Average=153.439 mg

Sr. No.	Tests	Specifications	Batch No. LT010E1001	Batch No. LT010E1002	Batch No. LT010E1003
5	**Content Uniformity**	AV NMT 15	AV:6.8	AV:4.0	AV:5.6
6	**Hardness**	NLT 3 Kp	Min= 5.823 kp Max=9.254 kp Average=7.896 kp	Min= 5.226 kp Max=9.981 kp Average=7.234 kp	Min= 5.320 kp Max=9.350 kp Average=7.986 kp
7	**Friability**	NMT 1.0% w/w	0.3% w/w	0.4% w/w	0.4% w/w
8	**Disintegration time**	NMT 30 min.	03 Min 18 Sec	03 Min 24 Sec	03 Min 27 Sec
9	**Dissolution**	NLT 75% (Q) of the labeled amount	Min=92% Max=100% Average=96%	Min=91% Max=100% Average=96%	Min=91% Max=104% Average=97%
10	**Assay**	95.0% to 105.0% of the stated amount	99.0%	99.0%	99.2%
11	**Microbial Limit Test**	TAMC: 1000 cfu/gm. TYMC: 100 cfu/gm. Escherichia coli : absent/gm. Salmonella : absent/10 gm. P. aeruginosa : absent/gm. S. aureus : absent/gm.	TAMC: 10 cfu/gm. TYMC: < 20 cfu/gm. Escherichia coli : absent/gm. Salmonella : absent/10 gm. P. aeruginosa : absent/gm. S. aureus : absent/gm.	TAMC: 10 cfu/gm. TYMC: < 20 cfu/gm. Escherichia coli : absent/gm. Salmonella : absent/10 gm. P. aeruginosa : absent/gm. S. aureus : absent/gm.	TAMC: 20 cfu/gm. TYMC: < 20 cfu/gm. Escherichia coli : absent/gm. Salmonella : absent/10 gm. P. aeruginosa : absent/gm. S. aureus : absent/gm.

19.0 Deviation and OOS:

There were no Deviation and OOS reported during entire validation activity

20.0 Change control:

There were no changes in process, procedure and specification during entire validation activity.

21.0 Summary Report:

Parameters are validated based on finding at each process.

Sr. No.	Process	Parameters	Validated parameters
1.	**Dry Mixing**	1. Mixing Time	10 Minutes
2.	**Wet granulation**	1. Amperage	18.85 A to 20.12 A
3.	**Drying of wet granules**	1. Drying Temperature (Inlet)	Between 50 to 70°C
		2. Drying Temperature (Outlet)	Between 35 to 50°C
		3. LOD	NMT 2%
4.	**Lubrication**	1. Speed of blender	08 RPM
		2. Mixing Time	5 Minutes
		3. Yield of lubricated granules	98.00% to 100.00%
		4. LOD	NMT 2%
5.	**Compression**	1. Speed of compression machine	16-42 RPM
		2. Yield of compression stage	98 to 100 %
6.	**Coating**	1. Inlet Air temp	60^0C to 80^0C
		2. Pan RPM	2-8 RPM
		3. Peristaltic pump speed	0.1 to 0.4
		4. Peristaltic pump speed(RPM)	5 to 40 RPM
		5. Yield of coating stage	97 to 100 %
7.	**Blister**	1. Packing Yield	99 %

22.0 Conclusion:

Process validation is carried out on the three standard batch sizes of 1,20,000 Tablets of product Solifenacin Succinate Tablets 5 Mg. During Process Validation Study all critical process parameters are validated and critical quality attributes are reviewed and found well within limits. Hence, process followed for manufacturing of Solifenacin Succinate Tablets 5 Mg stands validated.

These validation batches should be undertaken for stability study.

The validation batches should be put for stability studies in simulated packing.
Stability condition will be,

Condition	Storage	
	Temperature	**RH**
Accelerated	40°C ± 2°C	75% RH ± 5% RH
Long Term	25°C ± 2°C	60% RH ± 5% RH
Long Term/ Intermediate	30°C ± 2°C	75% RH ± 5% RH

23.0 Abbreviations:

LLS	:	Leben Life sciences
Mfg.	:	Manufacturing
Lic.	:	License
No.	:	Number
API	:	Active Pharmaceutical Ingredient
AR. No.	:	Analytical Report Number
Sr. No.	:	Serial Number
OFCB	:	Open Fronted Contentment Booth
HPLC	:	High Performance Liquid Chromatography
IR	:	Infra-Red
°C	:	Degree Celsius
%	:	Percent
LOD	:	Loss On Drying
FP	:	Finished Product
SOP	:	Standard Operating Procedure
QA	:	Quality Assurance
RH	:	Relative Humidity
Temp.	:	Temperature
BOM	:	Bill of Material
Min.	:	Minutes
MM	:	Millimeter
RPM	:	Rotation Per Minute
IPQA	:	In process Quality Assurance
Hrs.	:	Hours
NMT	:	Not More Than
KG	:	Kilogram
DT	:	Disintegration Time
QC	:	Quality Control
Av. Wt.	:	Average Weight
NLT	:	Not Less Than